
TENSOR NETWORK REPRESENTATION OF FOURIER TRANSFORMS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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ABSTRACT

The Fourier transform is represented by a tensor, when unfolding the vector transform using a binary coordinate representation. This tensor admits a sparse decomposition into (controlled) unitary matrices. We derive two standard algorithms in the contraction formalism:

- Fast Fourier Transform, by layerwise contraction of the tensor network
- Quantum Fourier Transform, by interpreting the tensor network as a quantum circuit

1 Unfolding

The Fourier transform of vectors in $\mathbb{C}^m$ is represented by a matrix

$$F[X = x, Y = y] = \frac{1}{\sqrt{m}} \exp \left[2\pi i \cdot \frac{x \cdot y}{m} \right]$$

where X, Y are variables of dimension m . The Fourier transform of a vector $V[X]$ is then the contraction

$$\langle V[X], F[X, Y] \rangle_{[Y]} .$$

We assume $m = 2^d$ and represent each index $x \in [2^d]$ by a tuple $(x_0, \dots, x_{d-1}) \in [2]^d$ such that

$$x = \sum_{k \in [d]} x_k \cdot 2^{d-k-1} .$$

Note that in this convention, x_{d-1} is the least significant bit. We follow Nielsen (2010) and denote x by $x_0 \dots x_{d-1}$. By $x_0 \dots x_{k-1} . x_k \dots x_{d-1}$ we denote $\frac{1}{2^{d-k-1}} \cdot x$.

The unfolded vector $V[X]$ is a tensor $V[X_{[d]}]$ with indices

$$V[X_{[d]} = x_{[d]}] = V[X] = \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1} \right) .$$

In the unfolded representation the fourier transform is described as the contraction with a tensor $F^d[X_{[d]}, Y_{[d]}]$ with coordinates

$$F^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \frac{1}{\sqrt{2^d}} \exp \left[2\pi i \cdot \left(\sum_{k \in [d]} x_k \cdot 2^{d-k-1} \right) \cdot \left(\sum_{k \in [d]} y_k \cdot 2^{d-k-1} \right) \right] ,$$

that is the Fourier transform of $V[X_{[d]}]$ is

$$\langle V[X_{[d]}], F[X_{[d]}, Y_{[d]}] \rangle_{[Y_{[d]}]} .$$

2 Tensor network decomposition

We now derive a tensor network decomposition of the tensor $F^d[X_{[d]}, Y_{[d]}]$.

Lemma 1. *The Fourier transform of basis vectors $\epsilon_{x_{[d]}}[X_{[d]}]$ is the elementary tensor*

$$\frac{1}{\sqrt{2^d}} \bigotimes_{k \in [d]} (\epsilon_0[Y_k] + \exp[2\pi i \cdot (0.x_{d-k-1} \cdots x_{d-1})] \cdot \epsilon_1[Y_k])$$

Proof. For each $y_{[d]}$ we have

$$\begin{aligned} & \langle \{(\epsilon_0[Y_k] + \exp[2\pi i \cdot 0.x_{d-k-1} \cdots x_{d-1}] \cdot \epsilon_1[Y_k]) : k \in [d]\} \cup \{\epsilon_{y_{[d]}}[Y_{[d]}}\} \rangle_{[\emptyset]} \\ &= \prod_{k \in [d]} \exp[2\pi i \cdot (0.x_{d-k-1} \cdots x_{d-1}) \cdot y_k] \\ &= \prod_{k \in [d]} \exp[2\pi i \cdot (x_0 \cdots x_{d-k-2} \cdot x_{d-k-1} \cdots x_{d-1}) \cdot y_k] \\ &= \exp \left[2\pi i \cdot x \sum_{k \in [d]} y_k \cdot 2^{d-k-1} \right] \\ &= \exp \left[2\pi i \cdot \frac{x \cdot y}{2^d} \right]. \end{aligned}$$

Since the basis coefficients of the elementary tensor coincide with those of the Fourier transform of $\epsilon_{x_{[d]}}[X_{[d]}]$, both tensors are equal. $\square$

We introduce the diagonal unitary

$$U^d[X_{[d]} = x_{[d]}, Y_{[d]} = y_{[d]}] = \begin{cases} \exp \left[2\pi i \cdot \left(\sum_{k \in [d]} x_k \cdot 2^{-k-2} \right) \right] & \text{if } x_{[d]} = y_{[d]} \\ 0 & \text{else} \end{cases}$$

Note that U^d is diagonal and the tensor product of matrices with variables X_k, Y_k . Diagonality can be used to contract U^d to an arbitrary tensor with computational demand in $\iota(2^d)$.

Using diagonality and elementarity, we have

Lemma 2. *The Fourier transform on $[2^d]$ can be decomposed into a Fourier transform on $[2^{(d-1)}]$ as:*

Proof. It suffices to show that the Fourier transform of arbitrary basis tensors $\epsilon_{x[d]} [X_{[d]}]$ follows the decomposition. The Fourier transform of $\epsilon_{[d-1]} [X_{[d-1]}]$ is by the Lemma above

$$\frac{1}{\sqrt{2^{d-1}}} \bigotimes_{k \in [d-1]} \left(\epsilon_0 [\tilde{Y}_k] + \exp [2\pi i \cdot (0.x_{d-k-2} \cdots x_{d-2})] \cdot \epsilon_1 [\tilde{Y}_k] \right)$$

In case of $x_{d-1} = 0$, the Fourier transform of $\epsilon_{x[d]} [X_{[d]}]$ is the tensor product of this state with $\epsilon^+ [X_0]$. This is reproduced by the decomposition, since the controlled unitary U^{d-1} acts trivially and the Hadamard gate transforms $\epsilon_0 [d-1]$ to $\epsilon^+ [X_0]$.

In case of $x_{d-1} = 1$, the leg vectors $\left(\epsilon_0 [\tilde{Y}_k] + \exp [2\pi i \cdot (0.x_{d-k-2} \cdots x_{d-2})] \cdot \epsilon_1 [\tilde{Y}_k] \right)$ are further transformed into

$$(\epsilon_0 [Y_{k+1}] + \exp [2\pi i \cdot (0.x_{d-k-2} \cdots x_{d-2}x_{d-1})] \cdot \epsilon_1 [Y_{k+1}]) .$$

In both cases we arrive at the Fourier transform of $\epsilon_{x[d]} [X_{[d]}]$. $\square$

By iteratively decomposing F^{d-1} until we arrive at F^1 (which is the H matrix), we get a decomposition of F^d into d H matrices and controlled U unitaries.

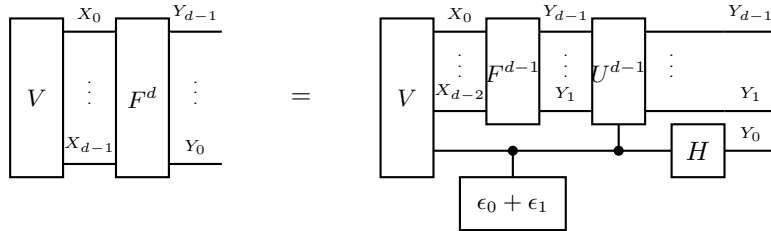
3 Fast Fourier Transform

3.1 Layerwise contraction

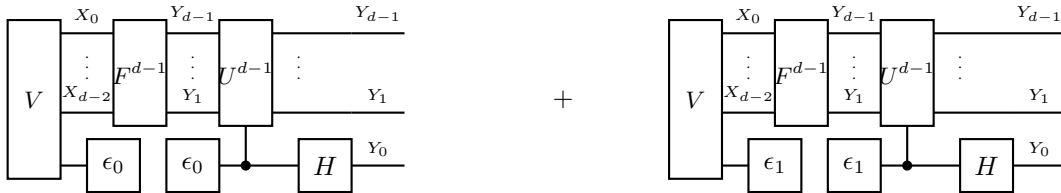
We contract the FFT tensor network layerwise. There are $2 \cdot d$ layers, each has a contraction demand in $\mathcal{O}(2^d)$ (exploiting diagonality of U). The algorithm has a total computational demand in $\mathcal{O}(d2^d)$.

3.2 Divide and conquer

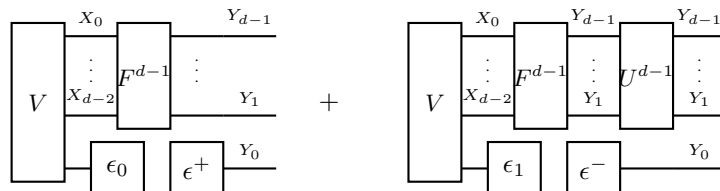
Called the radix-2 FFT Cooley and Tukey (1965). Can be seen as an iterative simplification by adding a $\mathbb{I} [X_{d-1}]$ and splitting it into the basis vectors.



Now by linearity of contractions, this splits into two terms



We simplify them to



The contraction of V with $\epsilon_1[X_{d-1}]$ respectively $\epsilon_1[X_{d-1}]$ are the even and the odd coordinates of V . Both are Fourier transformed with $d-1$ bits, which are further divided into Fourier transforms of lower order. In the conquer step of the FFT, the odd Fourier transform gets a coordinatewise transform by U , which is efficient since U is diagonal.

4 Quantum Fourier Transform

When $V[X_{[d]}]$ is a quantum state, the Fourier transform can be understood as a unitary evolving the state. Since both the controlled U tensor and the Hadamard H are unitary, each of them can be implemented by a sequence of gates. For sparse implementation one can exploit that U is a tensor product of matrices (i.e. single qubit gates).

To realize the QFT we need d single-qubit Hadamard gates and $\sum_{k \in [d]} k = \frac{d \cdot (d-1)}{2}$ controlled diagonal single-qubit gates (to realize the U^k transforms). The gate complexity is therefore $\mathcal{O}(d^2)$.

References

- James W. Cooley and John W. Tukey. An algorithm for the machine calculation of complex Fourier series. *Mathematics of Computation*, 19(90):297–301, 1965. ISSN 1088-6842, 0025-5718. doi: 10.1090/S0025-5718-1965-0178586-1. URL <https://www.ams.org/mcom/1965-19-090/S0025-5718-1965-0178586-1/>.
- Isaac L. Chuang Michael A. Nielsen. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, Cambridge ; New York, December 2010. ISBN 978-1-107-00217-3.